

Designing Virtual Work Environments for Knowledge Workers

Parisa Daeijavad*

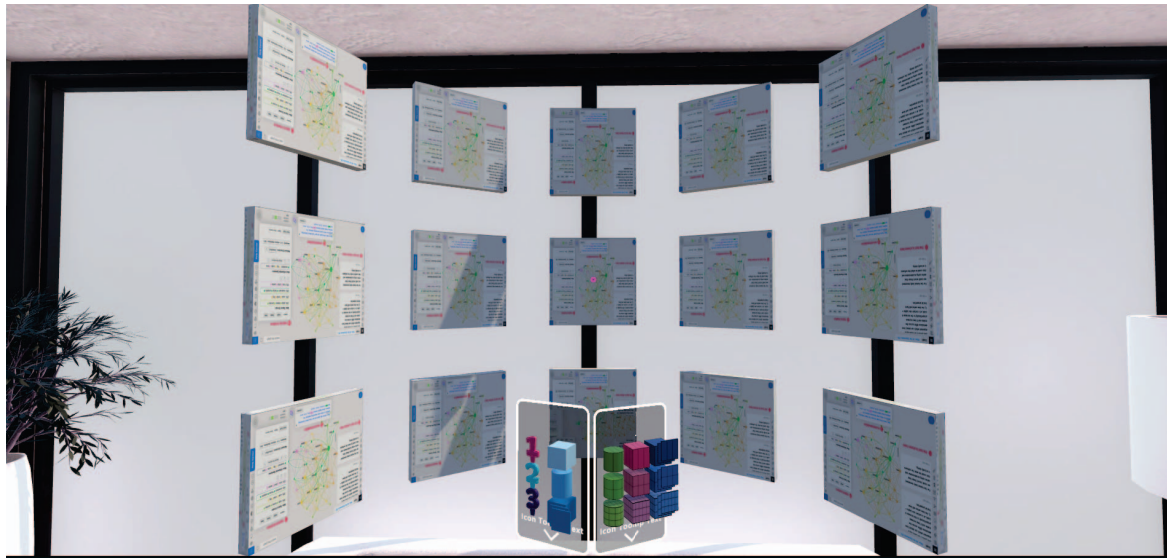
Nanjia Wang[†]Frank Maurer[‡]Department of Computer Science
University of Calgary

Figure 1: Immersive Virtual Work Environment: A curve of multiple displays that we designed for knowledge workers.

ABSTRACT

This position paper proposes the challenges faced by knowledge workers in current digital spaces and explores key design concepts for Extended Reality (XR) interfaces aimed at addressing these issues. Focusing on the unique needs of knowledge workers, we explore how XR technologies can ease current difficulties, providing a foundation for reshaping the future of work. The paper emphasizes the importance of spatial design considerations and user-centric approaches in developing XR interfaces that are suited to the complexities of knowledge work.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Interaction design—Interaction design process and methods—User interface design; Human-centered computing—Interaction design—Interaction design process and methods—User centered design; Human-centered computing—Human computer interaction (HCI)—Interaction techniques—Gestural input;

1 INTRODUCTION

In a variety of application domains, from game development [20] to medical [11] and industrial use [14], knowledge work has gained popularity in recent years. The term 'Knowledge Work' was initially

noted by Peter Drucker [7] where information workers (IWs) apply theoretical and analytical knowledge to develop products and services. A significant portion of knowledge workers' tasks may not involve physical papers, documents or objects. Instead, tasks are carried out using digital tools like laptops, tablets, or smartphones, all connected through the internet. This may enable knowledge workers to not limit their work location to a specific place.

Digital workspaces refer to specific digital business environments where digital technologies and information converge [19]. Unlike platforms designed for gaming or learning, digital workspaces are centred around various work scenarios [19]. The idea of a digital workspace has come up to adapt to the changes in how we use technology for work, to create a virtual space that supports the new ways we work.

With the widespread use of digital technologies, knowledge workers, who focus on tasks that demand theoretical and analytical thinking, often encounter challenges in the current digital setups [19]. As we delve into the world of digital workspaces, it's crucial to acknowledge these challenges faced by those involved in cognitive tasks. In today's world, Extended Reality (XR) is revolutionizing our digital spaces, bringing exciting opportunities for knowledge workers [19]. XR technologies go beyond just creating advanced digital work areas; they enable users to interact with digital elements and environments in a way that feels real [15]. In Figures 1 and 3, we illustrate the use of XR technologies in knowledge workers' workspace through our innovative design. Biener et al. in [4] utilized VR potentials for knowledge work, including extending 2D displays into the third dimension, selecting overlapping objects or displaying additional layers of information. These XR-driven digital workspaces not only immerse users in a more dynamic experience but also offer an engaging way to interact with digital content and make it easier for teams to have discussions about complex

*e-mail: parisa.daeijavad@ucalgary.ca

[†]e-mail: nanjia.wang1@ucalgary.ca

[‡]e-mail: frank.maurer@ucalgary.ca

ideas [18].

Although recently many papers focused on XR-driven digital workspaces and XR software [16] for knowledge workers, little research has been conducted to show digital workspace challenges and how XR can address these challenges. For instance, Beiner et al [3] focused on challenges that XR brings for knowledge work in everyday environments and proposed steps to address those open challenges. Wu et al [19] also discussed the current state and applications of XR in virtual teams in digital workspaces, however, they did not mention non-XR workspace challenges and how XR can improve such environments. In addition, there's a lack of evidence-based guidelines for designing XR-driven workspaces and guidelines are often researchers' opinions and ideas around the design and optimization of such environments. One example of this approach is the work done by Aufegger et al [1] where the authors observed knowledge workers to understand the potential for successful integration in VR and then evaluated environmental needs and opportunities through VR [1]. Our paper aims to highlight digital workspace challenges and state the potential for future work on developing evidence-based design principles for XR interfaces to enhance the efficiency and effectiveness of knowledge workers. By doing so, we envision a future where XR plays a role in reshaping the landscape of knowledge-intensive professions.

2 CHALLENGES IN CURRENT DIGITAL SPACES

Knowledge workers face limitations in digital environments, including spatial constraints, which often come from restricted square footage [13], lack of immersive collaboration, and difficulty representing complex information structures. The transition from physical to digital workspaces has not fully met the needs of knowledge-intensive tasks, hindering the seamless integration of digital technologies with the complexities inherent in knowledge-centric activities.

Ineffective Representation of Information The absence of a three-dimensional representation of information in traditional knowledge work environments makes it hard for IWs to perceive complex information. With traditional knowledge work environments, IWs need to analyze knowledge based on a 2D representation of information, lacking the more intuitive depth cues provided by a 3D representation. By incorporating some common intuitive depth cues such as perspective [10], 3D representations aim to simulate the visual experience of the real world, making it easier for users to understand and interpret complex information.

Spatial Constraints Traditional non-immersive 2D work environment poses a challenge for knowledge workers as it limits their ability to effectively utilize the workspace to organize and comprehend complex information fully. In these conventional settings, knowledge workers are limited to one static physical space and one or more digital devices with 2D interface. XR technologies offer a promising solution to enhance the knowledge work environment. VR enables knowledge workers to step into a virtual environment, breaking the spatial barriers of physical workspace with fewer distractions and more productivity [8]. On the other hand, AR overlays digital information onto the physical world, empowering knowledge workers to interact with data in real-time within their existing workspace [9]. This not only improves the efficiency of knowledge-intensive tasks but also enhances the overall work experience for individuals engaged in complex tasks [8].

Collaboration Limitations Traditional digital spaces often rely on utilizing 2D interfaces in a physical space. Such spaces lack the immersive collaborative experiences with interactivity necessary for effective collaboration. This makes collaboration difficult for knowledge workers in the traditional knowledge workspace setting. For instance, researchers have highlighted the issues of user proximity [12] and position arrangement of mobile devices [6] and

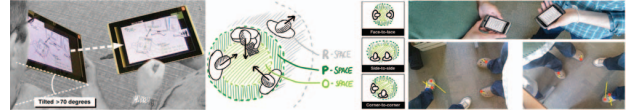


Figure 2: Tilting while pointing to an item on the display (left) [12], Devices shift in and out of the shared space (center) [12], orientation of each device relative to one another (right) [6].

tabletops [17] in physical collaboration workspaces (Fig. 2). These limitations can impact the natural flow of communication, and restrict shared understanding of complex information [5]. Using XR technologies with the shared view can improve task performance in collaborative workspaces [5].

Lack of smooth transition In traditional knowledge work environments, the disjointed experience arises from the insufficient integration between physical elements and digital interfaces. Users often need to transition seamlessly between tangible materials, physical documents, and digital content. The absence of smooth integration can be distracting, slowing down workflows and increasing cognitive load for knowledge workers. AR overlays digital information onto the physical world, allowing users to interact with both tangible and digital content simultaneously and VR provides immersive environments where knowledge workers can seamlessly navigate and manipulate digital content using controllers or hand-tracking.

3 KEY DESIGN CONCEPTS FOR XR INTERFACES

Transitioning from the challenges posed by current digital spaces for knowledge work, Extended Reality (XR) emerges as a potential solution that addresses these limitations comprehensively. As we explore the design principles in the following section, we will uncover how these principles provide the potential of XR to optimize knowledge work environments. Moreover, we will discuss the potential for future research in developing evidence-based design principles for XR interfaces, aiming to optimize task performance and enhance effectiveness for knowledge workers.

Improving Spatial Perception XR interfaces should enhance spatial understanding of information by improving depth cues. An XR-driven workspace will also provide a space where IWs can walk toward digital content to view detailed information from a closer perspective. In addition, if the workspace contains 3D visualization, walking inside the models can give better insights for further exploration. This addresses the challenge of spatial and information representation constraints, allowing knowledge workers to organize and manipulate complex data structures more intuitively.

Providing Large and Adaptive Workspace In comparison to traditional digital workspaces, XR-driven workspaces should notably increase the working space, allowing IWs to present digital content around themselves and making the comparison tasks much easier. With the additional space in XR, IWs can scale up all digital components, including displays, 2D digital documents and 3D models, even up to the size of an entire room to observe finer details. Furthermore, XR interfaces should offer adaptive workspaces for knowledge workers to customize virtual environments with their desired components such as desks, monitors, etc. based on their preferences and task requirements, addressing the challenges posed by static digital spaces.

Immersive Collaboration XR interfaces should facilitate immersive collaborative experiences, overcoming the current limitations of collaboration tools. For example, currently, IWs might communicate using voice and video. This experience can be enhanced by using avatars in a VR environment that convey the position and orientation of the user within their shared workspace to



Figure 3: Multiple displays in our virtual work design for knowledge workers.

others. Moreover, XR provides the opportunity for collaborators to share their perspectives compared to the situation where someone has to move beside another user to do so. This ensures a shared spatial presence, enabling knowledge workers to interact seamlessly with virtual content and remote colleagues.

Integration of Physical and Digital Elements To address the lack of seamless transition between physical content, and digital content, XR interfaces should enable the hands-on interactions for physical artifacts and digital information. XR controllers and hand tracking can be argued to be more efficient than mouse and keyboard interaction in knowledge-intensive tasks. In addition, it can be argued that XR can enhance the integration of physical and digital content by creating detailed visual effects and digital twins that mirror their real-world counterparts. For instance, Berkaoui et al. [2] placed 360° spheres in the scene to create a realistic representation of a real-world hall. Therefore XR technologies can facilitate the transition between contents for knowledge workers.

4 BUILDING A BASIS FOR FUTURE DISCUSSIONS

By addressing these challenges and proposing key design concepts, this paper aims to start discussions on the role of XR in reshaping the future of knowledge work. The presented key concepts serve as a starting point for considering how XR interfaces can be tailored to the unique needs of knowledge workers.

5 CONCLUSION

In conclusion, the challenges faced by knowledge workers in current digital spaces provide a strong motivation for exploring the potential of XR interfaces. By proposing key design concepts that directly address these challenges, we contribute to the dialogue on reshaping the future of work through extended reality. This paper encourages more exploration and discussion within the workshop, supporting a collaborative effort towards advancing XR technologies in knowledge-intensive professions.

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